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B.Tech. / M.Tech. (Integrated) DEGREE EXAMINATION, MAY 2025

Fourth Semester

21MAB204T - PROBABILITY AND QUEUEING THEORY

(For the candidates admitted during the academic year 2021 - 2022 to 2024 - 2025)

(Statistical table to be provided)

Note:

i. **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.

ii. **Part - B** and **Part - C** should be answered in answer booklet.

Time: 3 hours **Max. Marks: 75**

PART - A (20 × 1 = 20 Marks)

Answer **all** Questions

		Marks	BL	CO
1.	The first moment of a random variable X about origin is called (A) Variance (B) Mean (C) Standard Deviation (D) Covariance	1	1	1
2.	The conditions satisfied by the probability mass function $p(x)$ of a random variable X are (A) $p(x) \geq 0 \, \forall x \in X, \, \& \, \Sigma p(x) = 1$ (B) $p(x) \geq 0 \, \forall x \in X, \, \& \, \Sigma p(x) = 0$ (C) $p(x) \leq 0 \, \forall x \in X, \, \& \, \Sigma p(x) = 1$ (D) $p(x) \leq 0 \, \forall x \in X, \, \& \, \Sigma p(x) = 0$	1	1	1
3.	If X is a random variables then $Var(aX + b)$ is (A) $aVar(X) + b$ (B) $a^2Var(X) + b^2$ (C) $aVar(X)$ (D) $a^2Var(X)$	1	2	1
4.	The function real values which assign to every outcome of a sample space is called (A) random sample (B) random variable (C) random numbers (D) random experiment	1	1	1
5.	The moment generating function of a binomial random variable is (A) $(p + qe^t)^n$ (B) $(p + qe^{-t})^n$ (C) $(q + pe^t)^n$ (D) $(q + pe^{-t})^n$	1	1	2
6.	If the random variable X follows Poisson distribution with mean 3, then $P(X = 0)$ is (A) e^{-3} (B) e^{-2} (C) e^3 (D) e^2	1	2	2
7.	Normal distribution is the limiting form of _____ distribution under suitable statistical conditions (A) Uniform (B) Exponential (C) Geometric (D) Binomial	1	1	2
8.	The mean of the exponential distribution with probability density function $f(x) = \lambda e^{-\lambda x}, x \geq 0$ is (A) λ (B) λ^2 (C) $\frac{1}{\lambda}$ (D) $\frac{1}{\lambda^2}$	1	1	2
9.	If $P(X, Y)$ is a joint probability mass function of a two dimensional random variable (X, Y) then (A) $\sum_j \sum_i P(x_i, y_j) = 0$ (B) $\sum_j \sum_i P(x_i, y_j) = 1$			

(C) $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy = 0$ (D) $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy = 1$ 1 1 3

10. If X and Y are two independent continuous random variable then 1 2 3
(A) $f(x,y) = f_X(x) + f_Y(y)$ (B) $f(x,y) = f_X(x) - f_Y(y)$
(C) $f(x,y) = f_X(x)f_Y(y)$ (D) $f(x,y) = \frac{f_X(x)}{f_Y(y)}$

11. If mean = 2, standard deviation = $\sqrt{2}$ for each of n independent random variables with $n = 75$, then by Central limit theorem S_N follows 1 2 3
(A) $N(150, \sqrt{150})$ (B) $N(\sqrt{150}, 150)$
(C) $N(\sqrt{150}, \sqrt{150})$ (D) $N(150, 150)$

12. The marginal probability function of Y for the joint pdf $f_{XY}(x,y)$ is 1 1 3
(A) $\int\int f(x,y)dx$ (B) $\int\int f(x,y)dy$
(C) $\int f(x,y)dx$ (D) $\int f(x,y)dy$

13. In a queueing model $(a/b/c : d/e)$, d represents 1 1 4
(A) Queue discipline (B) System capacity
(C) Service time (D) Number of servers

14. In a queueing model $(M/M/1) : (\infty : FIFO)$, the probability of no customers (P_0) in the system is 1 1 4
(A) $\frac{\lambda}{\mu}$ (B) $\frac{\mu}{\lambda}$
(C) $1 - \frac{\lambda}{\mu}$ (D) $1 - \frac{\mu}{\lambda}$

15. Which term refers to 'a customer who leaves the queue because the queue is too long'? 1 1 4
(A) Balking (B) Reneging
(C) Jockeying (D) Leaving

16. In a shop with a single counter customers arrive at a rate of 20 per hour and are getting served at a rate of 30 per hour. The waiting time of a customer in the system is 1 2 4
(A) 2 min (B) 4 min
(C) 6 min (D) 8 min

17. A state i is an absorbing state if p_{ii} is 1 2 5
(A) 0 (B) 1
(C) 2 (D) ∞

18. The term ergodic means 1 2 5
(A) Irreducible and periodic (B) Irreducible and aperiodic
(C) Reducible and periodic (D) Reducible and aperiodic

19. The sum of all the elements of any row of the transition probability matrix is 1 1 5
(A) 0 (B) $\frac{1}{2}$
(C) $\frac{3}{4}$ (D) 1

20. The period d_i of any state i in a Markov chain is given by 1 2 5
(A) $gcd\{m, p_{ii}^m < 0\}$ (B) $gcd\{m, p_{ii}^m > 0\}$
(C) $gcd\{m, p_{ij}^m < 0\}$ (D) $gcd\{m, p_{ij}^m > 0\}$

PART - B (5 × 8 = 40 Marks) Marks BL CO

Answer **all** Questions

21. (a) A random variable X has the following probability distribution

8	3	1

X	0	1	2	3	4	5	6	7
P(X)	0	a	$2a$	$2a$	$3a$	a^2	$2a^2$	$7a^2+a$

Find (i) a (ii) P(1.5<X<4.5/X>2) (iii) Find the cumulative distribution function.

(OR)

(b) Let X be a random variable with $f(x) = e^{-x}$, $x \geq 0$, use Chebyshev's inequality prove that $P(|x-1| \geq 2) \leq \frac{1}{4}$. Also, show that the actual probability is e^{-3}

22. (a) In a normal distribution 31% of items are below 45 and 8% are over 64. Find the mean and standard deviation. 8 3 2

(OR)

(b) Derive moment generating function of a binomial random variable X and hence find its mean and variance.

23. (a) The joint probability density function of random variables X and Y is given by $f(x,y) = kxye^{-(x^2+y^2)}$, $x > 0$, $y > 0$. Find the value of k and also prove that X and Y are independent. 8 3 3

(OR)

(b) A fair coin is tossed 250 times. Find the probability that heads will appear between 120 and 140 times using central limit theorem.

24. (a) Customers arrive at a watch repair shop according to a Poisson process at a rate of one per every 10 minutes and the service time is exponential random variable with mean 8 minutes. 8 3 4
(i) Find the average number of customer in the shop
(ii) Find the average time a customer spends in the shop
(iii) Find the average number of customers in the queue
(iv) What is the probability that the server in idle?

(OR)

(b) A one-man barber shop can accomodate a maximum of 5 people at a time, 4 waiting and 1 getting hair cut. Customers arrive following Poisson distribution with average of 5 per hour and service is rendered according to exponential distribution at an average rate of 15 minutes.
(i) What is the percentage of idle time
(ii) Find the probability of a potential customer turned away
(iii) What is the expected number of customers in the queue
(iv) Find the expected time of a customer spent in the shop

25. (a) The one step transition probability matrix of a Markov chain $\{X_n\}, n = 0, 1, 2.....$ having state space $\{1, 2, 3\}$ is $\begin{bmatrix} 0.1 & 0.5 & 0.4 \\ 0.6 & 0.2 & 0.2 \\ 0.3 & 0.4 & 0.3 \end{bmatrix}$ and the initial distribution is $(0.7 \quad 0.2 \quad 0.1)$. Find 8 3 5

1. $P(X_2 = 3/X_0 = 1)$
2. $P(X_3 = 2, X_2 = 3, X_1 = 3, X_0 = 2)$
3. $P(X_2 = 3)$

(OR)

(b) A housewife buys 3 kinds of cereals A, B and C. She never buys the same cereal in successive weeks. If she buys cereal A, the next week she buys B. However, if she buys B or C, the next week she is 3 times as likely to buy A as the other cereal. In the long run, how often does she buy each of the three cereals?

PART - C (1 × 15 = 15 Marks)

Answer **any 1** Questions

		Marks	BL	CO
26.	There are three typists in an office. Each typist can type an average of 6 letters per hour. If letters arrive for being typed at the rate of 15 letters per hour. (a) What fraction of the time all the typists will be busy? (b) What is the average number of letters waiting to be typed? (c) What is the average time a letter has to spend for waiting and for being typed? (d) What is the probability that a letter will take longer than 20 min waiting to be typed and being typed?	15	4	4
27.	(a) The joint probability mass function of (X, Y) is given by $P(x, y) = k(2x + 3y)$, $x = 0, 1, 2; y = 1, 2, 3$ Find (i) k (ii) Marginal probability distributions of X and Y (iii) $P(X = x/Y = 1)$ (iv) $E(X), E(Y), Var(X), Var(Y)$ (v) $Cov(X, Y)$ (vi) Correlation coefficient between X and Y	15	4	3

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